

LAB REPORT
FOR
LAB - 1
EEC 556: SOFTWARE DEFINED RADIO

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Lab1 Part 1: This part of the Lab helps to understand the following palettes of Labview:

- Signal Processing Palette
- Digital Palette
- Structure Palette
- Complex Palette
- Array Palette

By replacing the Source and Error Detection VI files in given Block Diagram, simulator will allow to run multiple iteration and collect statistics from these runs.

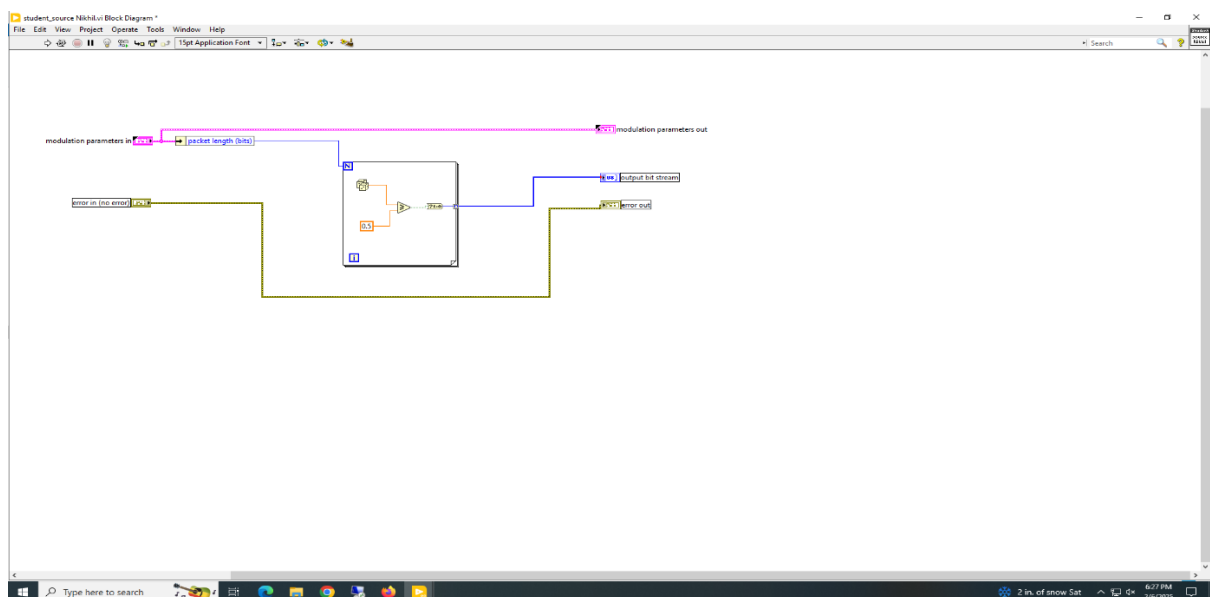


Figure 1: student_source Nikhil .vi (Block diagram)

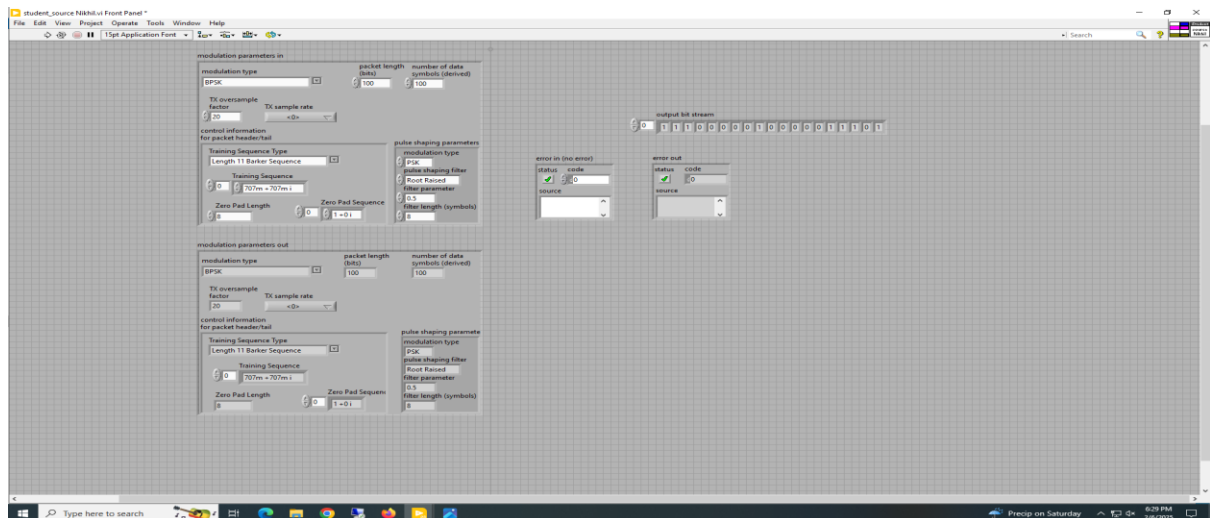


Figure 2: student_source Nikhil.vi (Front panel)

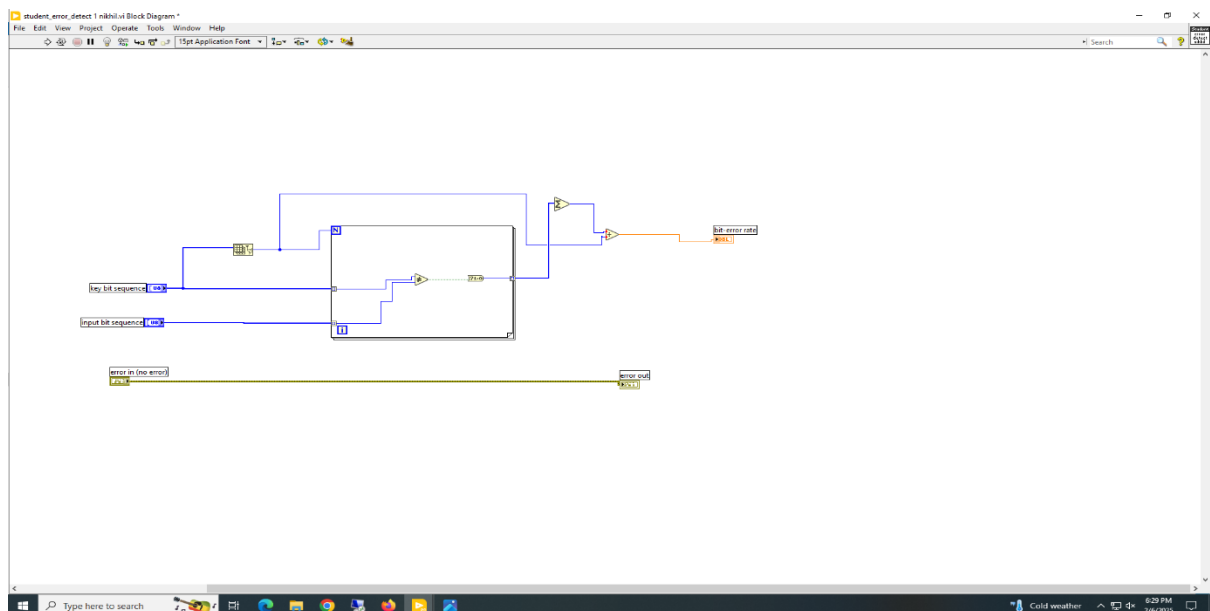


Figure 3: student_error_detect nikhil.vi(Block diagram)

After constructing student_source.vi (figure 1) and student_error_detect.vi (figure 3) with our own versions we have replaced them in awgn_simple_sim.vi (figure 4)

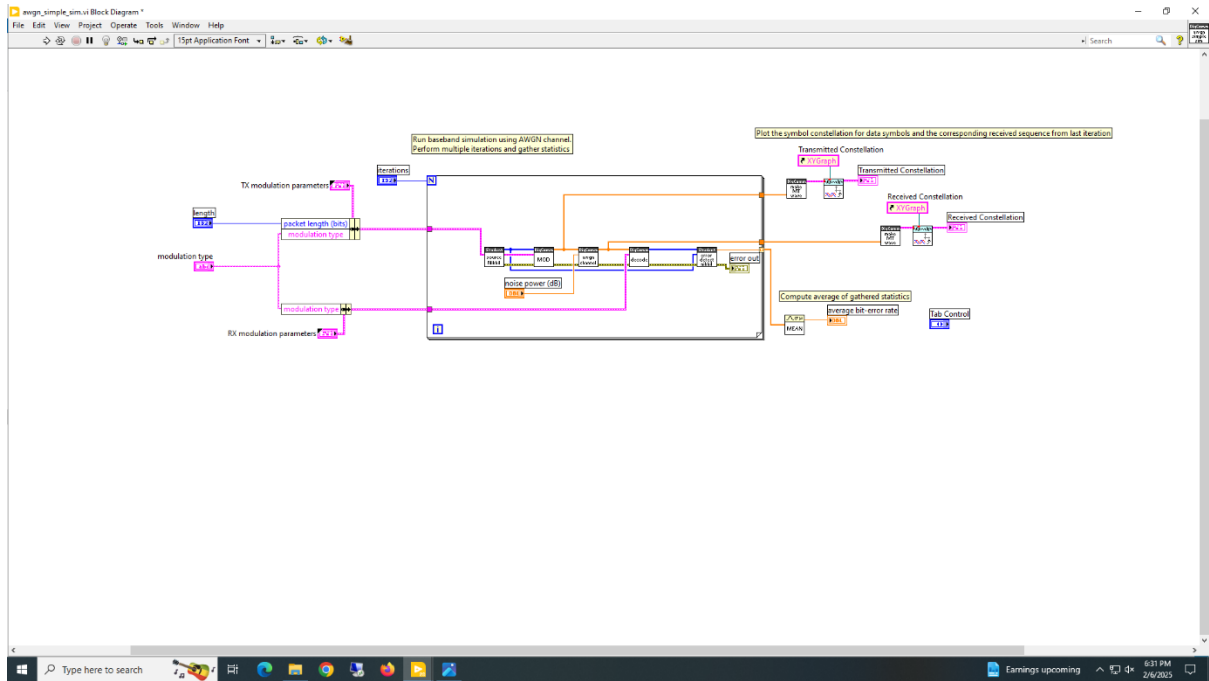


Figure 4: awgn_simple_sim.vi (Block diagram after replacing source and error detect)

After replacing them we have run the simulation on the front panel and observed the results as shown in below figure 5 and figure 6.

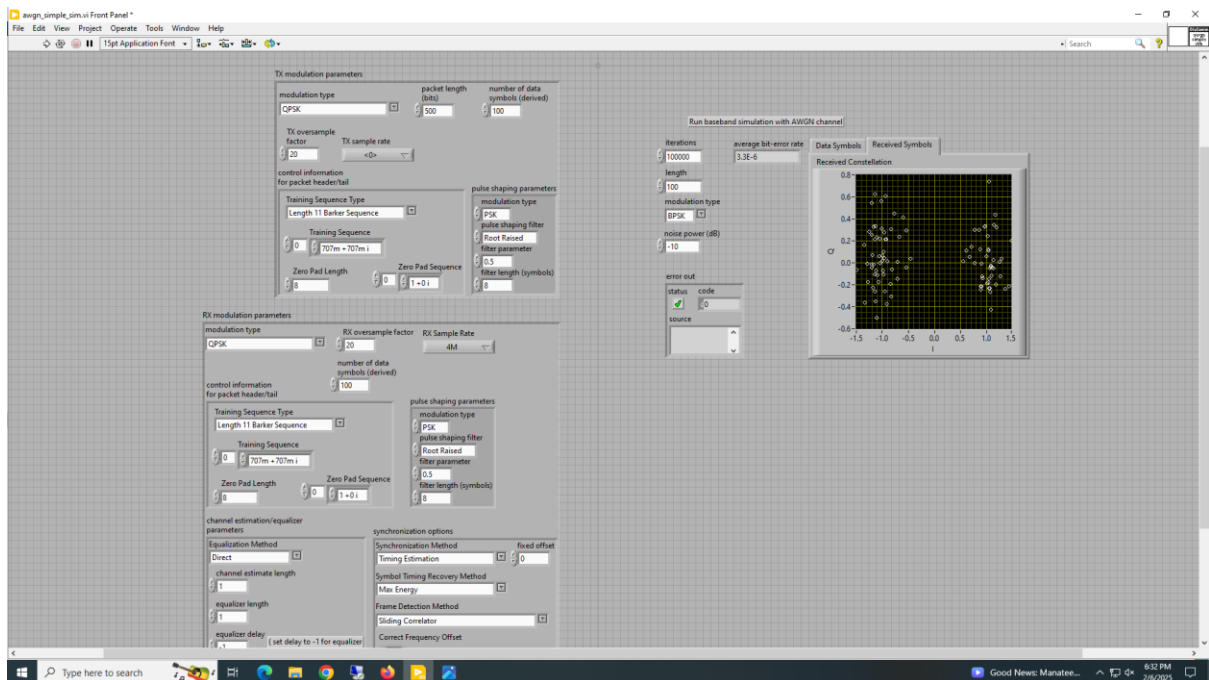


Figure 5: awgn_simple_sim.vi (Front panel 1)

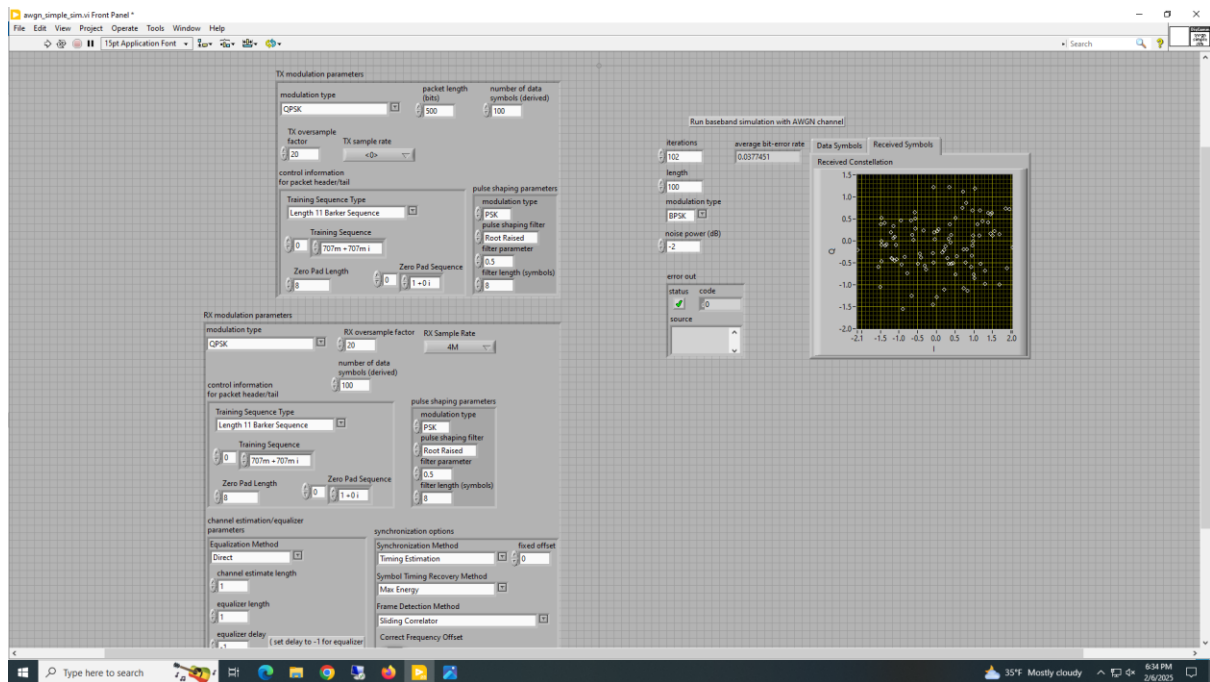


Figure 6: awgn_simple_sim.vi (Front panel 2)

I faced a problem with simulation part. To overcome this problem carefully read the manual and simulated the vi.

Lab1 Part2: Transmitter

The USRP receives a 16-bit resolution waveform from the host PC, sampled at rates up to 25 M Samples/sec. This signal is then upconverted to a radio frequency (RF), amplified, and transmitted over the air.

Building a Basic Sine Generation VI Build the function block in new VI using NI USRP TX palette.

After Following the Instruction given in the manual set the following parameter as below to generate the basic Sine wave.

The Function Block Diagram will look as following:

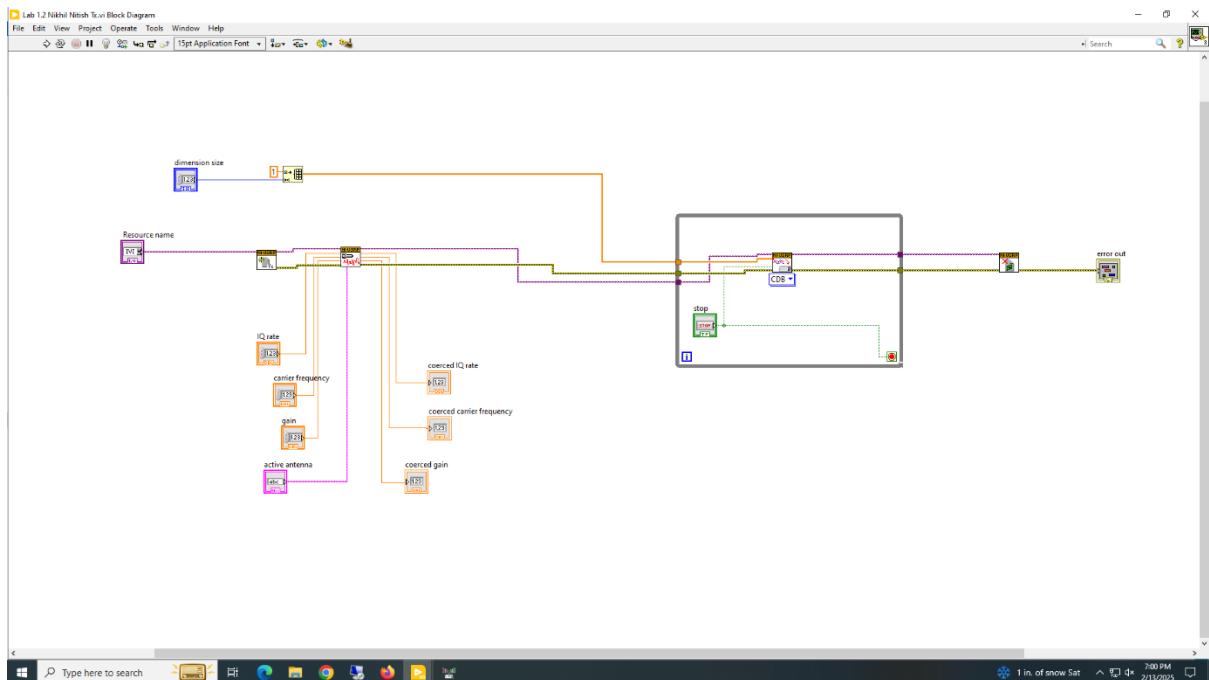


Figure 7: lab 1.2 nikhil nitish TX(Block diagram)

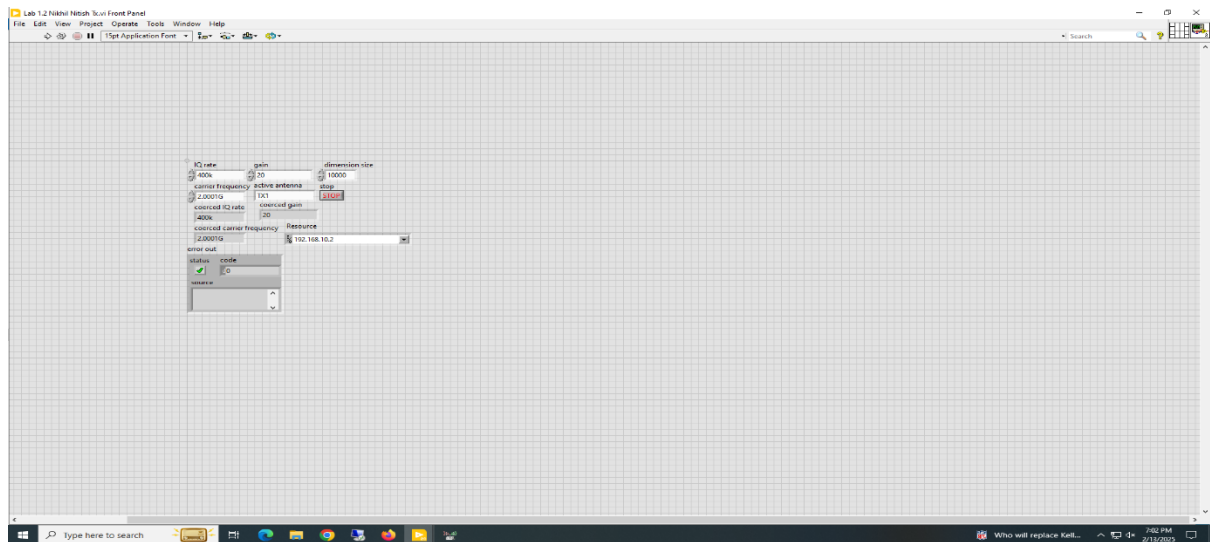


Figure 8: lab 1.2 nikhil nitish TX (Front panel)

On the Front Panel set the parameters as follow:

Carrier Frequency = 2.001 GHz

IQ Rate (Sample/Sec) = 400k

Gain (dB) = 20

Waveform Size = 10000

Active antenna = TX1

After following parameters as shown above, we have observed niUSRP EX spectral monitoring (interactive) on the new vi as shown in figure 9.

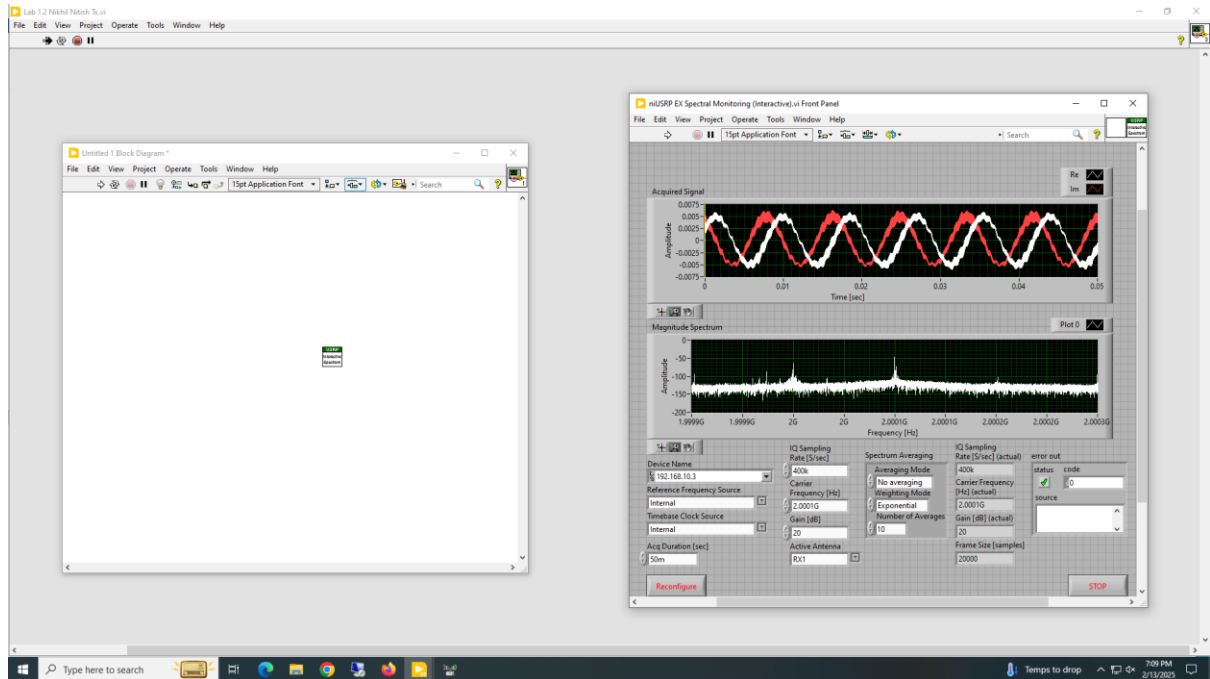


Figure 9: niUSRP EX spectral monitoring (interactive)

Lab1 Part2: Receiver

The USRP receives signals by mixing them with a desired carrier frequency to down-convert to a complex IQ baseband signal, which is then sampled at 100 M Samples/second. To reduce the required memory and processing load, the signal is downsampled to a user-specified rate. This decimation is crucial because sampling at the full digitizer rate (100 M Samples/sec) would capture a 20 MHz bandwidth, resulting in excessive data. By using a Digital Downconverter (DDC), the USRP can sample narrowband signals at much lower rates, typically just twice the signal bandwidth, reducing memory requirements.

Now following the process available in manual Block diagram can be constructed for receiver as below.

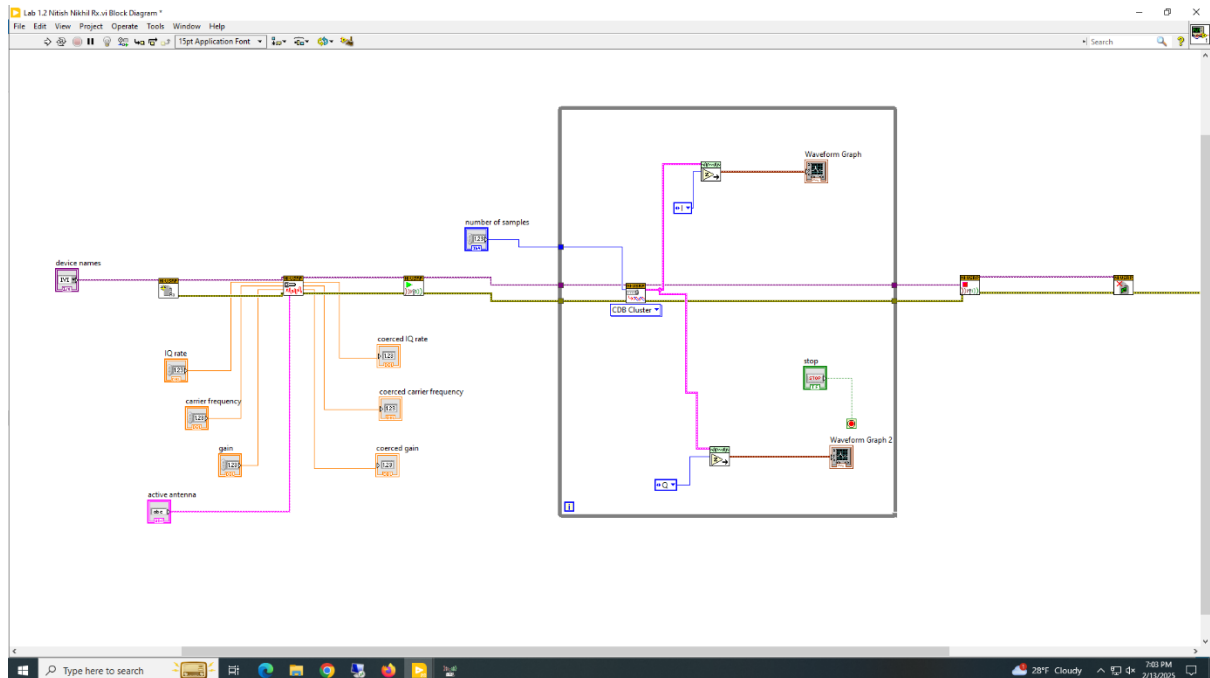


Figure 10: lab 1.2 Nikhil Nithish RX.vi(Block diagram

We have Connected both USRP devices with PC and run both VIs to generate Continuous Sine wave signal.

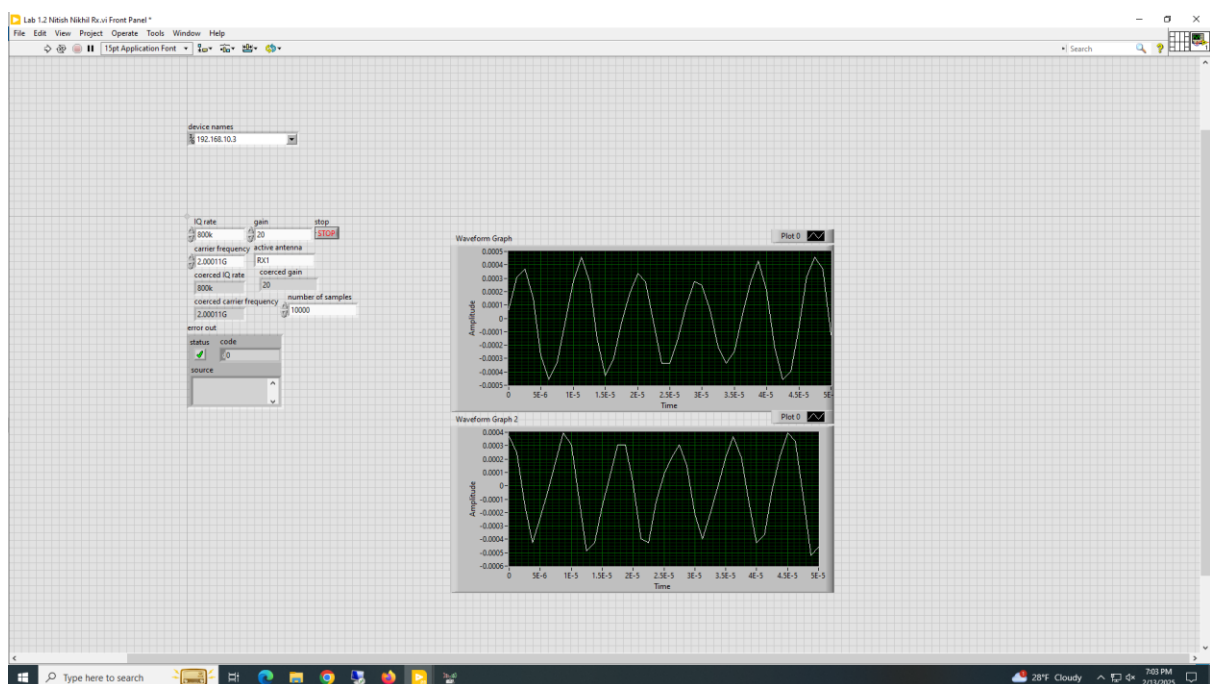


Figure 11: Lab 1.2 Nikhil Nithish RX.vi(Front panel)

As we can observe some noise and other factors like transmission through air are affecting received sine wave in figure 11.

On the Front Panel set the parameters as follow:

Carrier Frequency = 2.00011 GHz

IQ Rate (Sample/Sec) = 800k

Gain (dB) = 20

Waveform Size = 10000

Active antenna = RX1

Questions and answers

- What is the Range of allowable carrier/center frequency supported by the NI-USRP?

Ans: Device used in Lab is USRP-2920 which support the range of 50 MHz to 2.4 GHz.

- What is the Maximum allowable bandwidth supported by the NI-USRP?

Ans: 25MHz for USRP-2920

- What is the Maximum sampling rate of the NI-USRP?

Ans: 25M Samples/Sec for USRP-2920 •

- Why do you think the DDC is implemented? What is its main benefit?

Ans: Reducing the sampling rate of a discrete-time signal is essential because if the ADC samples at the full digitizer rate (100 M Samples/sec), it can capture a 20 MHz bandwidth, leading to a large amount of memory usage. Decimation is therefore necessary. By using the Digital Downconverter (DDC), the USRP can acquire narrowband signals at a much lower sampling rate than the full digitizer rate, significantly reducing the memory required to store the waveform.

- In your own words, describe what the bandwidth of an instrument is.

Ans: Bandwidth refers to the analog front end's capacity to transfer a signal from the external source to the ADC with minimal amplitude loss, from the probe or test fixture to the ADC input. It is defined as the frequency at which a sinusoidal input signal is attenuated to 70.7% of its original amplitude, also known as the -3 dB point. Bandwidth is measured between the lower and upper frequency points where the signal amplitude falls below the passband. It is calculated as $BW = f_2 - f_1$, where f_1 and f_2 are the corner frequencies.

- What is meant by the sampling rate of an instrument?

Ans: The Sampling rate is the frequency at which the ADC converts the analog input waveform to digital data. The higher the sampling rate the greater the resolution and details can be seen in the waveform and vice versa.

- Why are these specifications important for designing a transmitter and receiver in a wireless communications system?

Ans: Both parameters are crucial in designing a modulation system. As explained above, bandwidth represents the maximum amount of data that can be transmitted over a channel in a given time period; thus, a higher bandwidth results in greater data capacity. On the other hand, the sampling rate determines the quality or resolution of the received signal. A higher sampling rate leads to better resolution and quality of the output. To accurately reconstruct the signal, the sampling rate should be at least twice the signal's bandwidth.

- What is the IP Address for your NI USRP.

Ans: IP Addresses of NI USRP are for TX: 192.168.10.2 & RX: 192.168.10.3

- Describe what happens to the spectral response of the transmitted signal when you place an obstruction between the antenna of transmitter and that of receiver (i.e. blocking the line of-sight component of the waveform).

Ans: Obstruction placed between the antennas causes distortion of signals responses. Noise, interference added into the signals may result into weak signal or signal accompany by noises at receiving end.

- What are the three elements in the data cluster output from niUSRP Fetch Rx Data (Poly).vi? Give the name and data type of each element.

Ans: Three elements in the data cluster output from niUSRP Fetch Rx Data (Poly).vi are as follow

Data Cluster element name	Description	Data Type
t0	trigger/start time of the acquired Y array	Double 64-bit real
dt	the time between values in Y array	Double 64-bit real
Y	the complex value baseband waveform	Double complex

- In RXRF_config.vi the IQ rate parameter is used by niUSRP ConfigureSignal.vi. Can the USRP support any arbitrary IQ rate? If not, what will the USRP do when an unsupported bandwidth parameter is requested?

Ans: No it will not support any arbitrary IQ rate. USRP device will coerce/adjust parameter i.e. the IQ rate, supported by USRP device.